

FEMINISATION OF AGRICULTURAL WORKFORCE IN INDIA

(A dissertation submitted for the partial fulfillment of award of
the Bachelors' degree in Economics)

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DECLARATION

I, Smitashree Das do hereby declare that this report is an original work of my Dissertation. It has been written by me and has not been submitted for any other previous degrees. The collaborative contributions have been indicated clearly and acknowledged. I declare that this project report is prepared by myself under the guidance of Dr. Swatilekha Thakur.

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CERTIFICATE

This is to certify that the current dissertation “Feminisation of Agricultural Workforce in India” is submitted by Smitashree Das (21116013), KIIT school of Social, Financial and Human Sciences, KIIT Deemed to be University. It is a record of confide work carried out by her during her course period under my supervision and guidance.

It embodies the report of her qualitative research study on Feminisation in Agricultural Sector and has not been submitted to any other University for the award of any degree.

I wish her all the best and success in future endeavors.

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ABSTRACT

The agricultural sector in India has witnessed a significant increase in the participation of women in various roles such as cultivators, entrepreneurs, and labourers, leading to a phenomenon termed as the 'feminization' of agriculture. Despite their substantial contributions, women's role in agriculture has often been overlooked in policy frameworks and official statistics. This research dissertation aims to shed light on the gender dynamics within the agricultural workforce in India, focusing on female participation rates, wage disparities, and the factors influencing female wage rates. The study utilizes secondary data from various sources to analyze trends in female participation rates and wage differentials over time. Additionally, a regression analysis is conducted to examine the relationship between average female wage rates and variables such as state GDP, rural female literacy rate, and female labour force participation rate in rural areas. The findings of this study contribute to a better understanding of the socio-economic factors influencing female employment in agriculture and provide insights for policy interventions aimed at promoting gender equity and economic empowerment in the agricultural sector.

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CHAPTER 1 : INTRODUCTION

There is ‘feminisation’ of agriculture sector, with number of women increasing in multiple roles as cultivators, entrepreneurs and labourers. The success in agriculture triggered a deterministic transformation of a fairly diverse landscape of the country and from the 1960s to this date, India witnessed a blend of initiatives (public and private) shaping the agriculture sector and reinforcing the ‘green revolution’ pathway. However, there is one significant part of the agricultural revolution, which continues to be disregarded across the country, it is the women’s significant contribution to agriculture in rural India. It is known that 78 per cent of India’s employed women work in agriculture. As per the Annual Periodic Labour Force Survey, 2021-2022, agriculture has the highest estimated female labour force participation of 62.9 per cent. The rural women in India are contributing to this sector for many years, but mostly as labourers in their own farms and not “farmers”. The Agriculture Census (2010-11) shows that out of an estimated 118.7 million cultivators, 30.3 per cent were females. Similarly, out of an estimated 144.3 million agricultural labourers, 42.6 per cent were females. According to Census 2011, there has been a 24 per cent increase in the number of female agricultural labourers between 2001 and 2011. As per the Census 2011, out of the total female main workers, 55 per cent were agricultural labourers and 24 per cent were cultivators. Economic Survey 2017-18, says with growing rural-to-urban migration by men, there is ‘feminisation’ of the agriculture sector, with an increasing number of women in multiple roles as cultivators, entrepreneurs, and labourers. With the participation of women in agriculture and allied activities becoming more significant, it becomes essential to keep women at the center of India’s policy initiative.

Women form an integral part of the Indian workforce. The total number of female workers in India is 149.9 million and female workers in rural and urban areas are 121.9 and 28.0 million respectively (source:census,2011). Out of total 149.9 million female workers, 35.9 million females are working as cultivators and another 61.5 million are agricultural labourers”. Of the remaining female’s workers, 8.5 million are in household Industry and 43.7 million are classified as other workers

The growing recognition that women can become an agent of economic and social transformation has also started gaining wide acceptance in the policy and academic circles. The women's participation as workforce plays a major role in determining women's autonomy and self-dependency. As opined by famous economist Amartya Sen that women agency should become a determinant factor to take care of overall development of society. In nations with a high human development index, females are more consistently employed in every sector at par with their male counterparts. In our country, the Female Labour Force Participation (FLEP) rate has remained visibly low and the International Labour organization ranks India's FLEP rate at 121 out of 131 countries in 2013, one of the lowest in the world. India has the lowest FLEP rate in South Asia, except for Pakistan. The agriculture sector is the backbone and livelihood of more than 50% of overall manpower of India. where women play the role of a key partner. Women constituted 38% of the agricultural labour force and employ two-third of the female workforce in India. Women are working extensively in farm activities like production of major grains and millets, sowing, applying manure, fertilizer, and pesticide, seed selection and seedling production, weeding, transplanting, threshing, winnowing and harvesting, in livestock production, fish processing, collection of non-timber forest produce (NTFP), etc. Also, the government has moulded an agricultural policy of India by integrating women as active agents in rural transformation

The history of the feminization of the agricultural workforce in India is deeply intertwined with the country's socio-economic and cultural evolution. While women have always played essential roles in agriculture, their contributions were often overlooked or undervalued in official statistics and policy frameworks. Historically, women in India have been actively involved in various agricultural activities, including planting, weeding, harvesting, and post-harvest processing. These tasks were often considered part of women's domestic responsibilities and were not formally recognized as part of the agricultural workforce. During the colonial era, agricultural policies and land tenure systems marginalized women's access to land and resources, further restricting their participation in formal agricultural activities. However, women continued to contribute to agricultural production through informal and unpaid labor within household farms. After independence, efforts to promote rural development and agricultural modernization began to recognize the importance

of women's contributions to the agricultural sector. However, gender disparities persisted, with women facing barriers to land ownership, access to credit, extension services, and technology. The latter half of the 20th century saw the emergence of women's movements and advocacy groups that highlighted the need to address gender disparities in agriculture. These movements advocated for policies and programs aimed at empowering women farmers, improving their access to resources, and recognizing their contributions to agricultural development. In response to these advocacy efforts, the Indian government introduced various schemes and programs aimed at promoting women's participation in agriculture. These initiatives included the formation of self-help groups, training programs, financial assistance, and support for women-led agricultural enterprises. In the 21st century, there has been a growing recognition of the feminization of the agricultural workforce in India. Women's participation in agricultural activities, both as unpaid family laborers and as wage workers, has increased significantly. Government policies and programs continue to focus on empowering women in agriculture and addressing gender disparities in access to resources and opportunities.

Overall, the history of the feminization of the agricultural workforce in India reflects a complex interplay of socio-economic, political, and cultural factors. While progress has been made in recognizing and supporting women's contributions to agriculture, challenges such as gender disparities in land ownership, access to resources, and decision-making continue to persist and require ongoing attention and intervention.

The gender wage gap, the difference between male and female average wage rate has long been noted and debated in the Indian agricultural labor market. It is observed that the female agricultural laborers do not enjoy an equal or equivalent wage rate for the same or equivalent work as compared to the male agricultural laborers. In 1976, the Equal Remuneration Act was adopted for equal pay both of male and female agricultural laborers for the "same work or work of a similar nature" to protect the rights of the female laborers against unfair wage practices. Still, it is observed that the female wage rates are not yet equal or equivalent to the male wage rates. Instead, an absolute gender wage disparity has been observed over time. As per census 2011, among the total rural workforce, female laborers classified as farm laborers is 38.9 percent compared to the 20.8 percent male agricultural laborers.

Mencher and Saradamoni(1982) have observed that the female laborers, mainly among the landless (or marginal farmer) households contribute more than half (or close to half) of the total family income and most importantly without the women's income; these households might not be able to fulfill their basic needs. Despite performing similar work under the same working condition, the female laborers are mostly unorganized and unaware of their constitutional rights. Therefore, more than 90 percent of the rural females are treated as a cheap and secondary source of laborers (Javeed &Manuhaar, 2013). In Indian agricultural labor market, it is believed that female agricultural laborers cannot do heavy work due to their low muscle strength and malnutrition. It is often argued that the male laborers due to their greater physical strength and energy are more productive and efficient than the female laborers and hence from the point of view of the employer; the male laborers deserve to have better wage rates and more person-days compared to the female laborers even for the same agricultural task (Kundu, 2013). It is observed in every state of India that the female agricultural laborers are not employed during the time of ploughing. Even for the gender-neutral tasks such as weeding, sowing, transplanting, and harvesting, the female laborers are still largely discriminated from the male laborers in terms of wage rates.

CHAPTER 2 : LITERATURE REVIEW

Sen (1999), in "Development as Freedom," emphasizes the potential of women to drive social transformation. He argues that women's well-being depends on factors like their ability to earn independent income, access employment opportunities, own property, and attain higher social status.

Deere (2005) scrutinized census data from Latin America and identified inherent flaws in census methodologies. These flaws primarily stem from the tendency to conform to socially prescribed gender roles, portraying women as primarily engaged in household duties rather than accurately reflecting their economic activities. The focus on income-generating activities overlooks women's contributions in subsistence production, such as livestock care, post-harvest processing, and tending to kitchen gardens. Additionally, census questions typically fail to capture the seasonality of agricultural work and women's off-farm employment.

Bina Agarwal (2008) *Women, poverty and agricultural growth in India*, This article delves into the connections among gender, poverty, and agricultural development in India. It highlights how women and girls from impoverished rural households disproportionately bear the brunt of poverty, particularly evident in biased practices within households regarding food and healthcare allocation. Notably, there are significant variations across regions, with the bias being more pronounced in the northwestern states compared to the southern ones. The article discusses potential economic, social, and historical factors contributing to these regional differences. It also addresses the specific challenges faced by households headed by women. Furthermore, it critically evaluates the ongoing discourse on the correlation between rural poverty and agricultural advancement. Additionally, the article conducts a comprehensive quantitative analysis of how the adoption of new agricultural technologies and ensuing growth impact the employment and earnings of both female and male agricultural laborers, who form the majority of the rural poor. It explores the link between changes in these economic indicators and other factors, such as shifts in dowry practices, and their implications for the situation of impoverished rural women across different geographical regions.

Sudaeshan & Bhattacharya (2009) study women's workforce participation in Urban Delhi, finding that it exceeds previous estimates. They explore various factors influencing women's decisions to engage in paid work outside the home, such as education, domestic responsibilities, and safety concerns in public spaces.

Dewan, Ritu(2016) Contextualising and Visibilising Gender and Work in Rural India: Economic Contribution of Women in Agriculture The paper aims to explore the intricate relationship between gender, poverty, and agricultural growth in India. It seeks to shed light on how women and female children from impoverished rural households bear a disproportionate burden of poverty, particularly evident in biased practices within households regarding food and healthcare allocation. The objective is to examine the dynamics of women's labor and livelihoods, especially in rural settings, considering various economic, social, and historical factors. Methodologically, the paper employs a qualitative approach, analyzing existing data and literature to understand the complexities of gender, poverty, and agricultural development. The time period under consideration spans several decades, allowing for a comprehensive analysis of trends and changes. The analysis reveals significant cross-regional variations in the extent of bias against women, with the northwestern states exhibiting higher levels of discrimination compared to the southern ones. The paper discusses potential economic, social, and historical factors contributing to these regional differences, as well as the specific challenges faced by female-headed households. Furthermore, the paper critically examines the ongoing debate on the relationship between rural poverty and agricultural growth. It conducts a detailed quantitative analysis of the differential effects of new agricultural technology on the employment and earnings of female and male agricultural laborers. Additionally, the implications of changes in economic variables on issues such as dowry payments are explored. The findings highlight the centrality of gender equality in addressing poverty and promoting sustainable livelihoods, particularly through ensuring women's rights to own and control resources, including land. Policy initiatives aimed at enhancing women's land rights have shown promising results, with an increase in the share of women landholders over time.

Fox and Pimhidzai (2013) analyzed census data from Uganda and revealed a systemic underestimation of women's labor force participation due to the framing of labor as

predominantly conducted outside the home. This framing is inconsistent with the reality that many economic tasks are performed within the homestead by women.

Ghosh & Ghosh (2014) discuss the fluctuating participation of women in agriculture between 1961 and 2001, attributing it to discriminatory practices. This piece of research is highlighting the trend of female participation in agriculture across various Indian states. The secondary data collated for the research is used to study the growth trend of the agricultural worker from 1961-2001. Analysis is done to categorize the states on their identical behaviour of participation in agriculture by hierarchical clustering of economically active female in agriculture based on measurements like coefficient of variation, compound growth rate and work participation rate. They note a shift of female labor from agricultural to non-agricultural sectors.

Tamang et al. (2014) employed a literature review to inform their qualitative case study approach on trends in the feminization of agriculture. Baada and Najjar (2020) similarly utilized literature reviews to investigate women's increasing involvement in agricultural labor due to male outmigration, particularly in drylands. However, they noted a lack of research on migration-related feminization of agriculture and its impact on food security and productivity.

Swamikannan & Jeyalakshmi (2015) examine the trends in female involvement in agriculture across different Indian states, analyzing data from 1961 to 2001. The objectives of the present study are (i) To measure the season wise employment of woman labour in agriculture, (ii) To examine the distribution of woman workers in India. The study is based on only secondary data which were collected from books, journals, government reports, websites and NSSO data. They find that women encounter discrimination in agrarian societies, with their status consistently lower than that of male counterparts. Government interventions aim to address this disparity by amending laws.

Zhang et al. (2015) combined various datasets to challenge the notion of the feminization of agriculture in China, suggesting that younger women's out-migration from farming contributes to a more balanced gender distribution in agricultural activities.

Tumbe (2015) and Ambler et al. (2020) also raised concerns about the accuracy of census data, suggesting that fatigue and biases among respondents could lead to data losses and distortions.

Itishree Pattnaik et.al (2017) in her paper *The feminization of agriculture or the feminization of agrarian distress? Tracking the trajectory of women in agriculture in India* The objective of this paper is to examine the phenomenon of the feminization of agriculture in India and its implications for the changing dynamics of rural life, especially regarding women's social and economic roles. The methodology involves analyzing four sets of occupational data obtained from the Indian Census for the years 1981, 1991, 2001, and 2011. The time period under study spans three decades, allowing for a longitudinal assessment of the trend. The analysis reveals that the increasing involvement of women in farm work, driven primarily by the outmigration of men from rural areas, does not necessarily correlate with broader indicators of women's social or economic empowerment. Instead, it is closely linked to various poverty-related factors. The findings suggest that women's heightened participation in agriculture exacerbates the already substantial workload borne by rural women, thereby exacerbating their overall well-being.

Mahata, Kumar, & Kumar (2017) analyze the regional disparities in literacy rates, sex ratios, and women's workforce participation in Haryana, highlighting the need for social and governmental interventions to improve women's empowerment.

Descriptive statistics are frequently employed to contextualize smaller qualitative studies, as demonstrated by Gartaula and Niehof (2018), who utilized surveys and qualitative case studies to examine the impact of male outmigration on land and farming dynamics in Nepal.

Nitya Chanana-Nag & Pramod K. Aggarwal(2018) *Woman in agriculture, and climate risks: hotspots for development* The paper aims to provide impact-driven solutions in the intersection of gender and climate change adaptation, reflecting the growing interest among research and development practitioners. It proposes a methodology to identify hotspots where interventions targeting gender-specific needs in climate

change adaptation could be prioritized, with a focus on women farmers in India. The study period covers recent years to capture contemporary climate trends and socio-economic conditions. The methodology entails linking climate change risks experienced by women farmers with their social profiles and agricultural roles. By analyzing climatic data and socio-economic indicators, the study identifies 36 hotspots across 10 states in India where women farmers are significantly affected by high probabilities of drought, excess rainfall, and heatwaves. These hotspots represent around 14.4% of the total women farmers in the country. Furthermore, the paper conducts a socio-economic characterization of the hotspot populations, revealing barriers such as limited access to labor, credit, and markets for female cultivators, along with lower wage rates for female laborers. Based on these constraints and the climatic risks faced by women in the hotspots, the study emphasizes the potential of climate-smart agriculture technologies and practices. Additionally, the paper compares its findings with current research in the field, highlighting the potential to leverage existing efforts for the efficient scalability of gender-sensitive climate change adaptation interventions.

Spangler and Christie (2019) conducted feminist political ecology studies to explore decision-making processes in male outmigration households, revealing nuanced power dynamics and challenges in drawing overarching conclusions about women's empowerment.

Mamata Swain et al. (2019) in the paper *Feminisation of Agricultural Labour in India: Issues and Challenges*. In their paper titled "Feminisation of Agricultural Labour in India: Issues and Challenges," Mamata Swain et al. (2019) explore the increasing involvement of women in agricultural work in India and the associated issues. They aim to review and understand the challenges related to the feminisation of agriculture, focusing on changes in the nature of women's work, gender-based wage differentials, and implications for women's empowerment. The methodology employed involves a comprehensive review of existing literature in agricultural economics and social anthropology. The authors analyze trends over time, drawing on data from various sources to examine the evolving role of women in agriculture. Over the period from January to March 2019, the authors conducted an in-depth analysis of the literature and synthesized key findings to present a nuanced understanding of the feminisation

of agricultural labour in India. In their analysis, Mamata Swain et al. found that women's participation in agricultural work has increased over the years, leading to a feminisation of the agricultural workforce. However, this trend is accompanied by gender disparities in wages, with women typically receiving lower pay than men for similar work. The authors identify various factors contributing to these wage differentials, including productivity differences, labour market segmentation, social norms, and lack of bargaining power among women. The authors conclude by suggesting policy implications to address the challenges identified. They advocate for capacity building among women through education and training, public investments in infrastructure to reduce domestic burdens, and the development of women-friendly agricultural technologies. Additionally, they emphasize the need for effective extension services to disseminate information and narrow gender wage differentials.

Innovative methods, such as aspirational narratives deployed by Crossland et al. (2021) in Kenya, are emerging to better understand the implications of the feminization of agriculture. These methods, like SenseMaker®, utilize narrative-based survey tools complemented by focus group discussions and semi-structured interviews to capture nuanced insights.

Nozomi Kawarazuka et al. (2022) introduces the paper which aims to challenge common myths surrounding the concept of the "feminization of agriculture" through a literature review. The objective is to analyze and debunk four prevailing myths regarding this phenomenon and its implications for global food security. The methodology involves examining existing literature to unravel each myth and provide evidence to counter them, thereby offering a more nuanced understanding of gendered power dynamics in agriculture. The time period covered in the analysis spans various studies and data sources contributing to the discourse on feminization of agriculture. The analysis begins by challenging the myth that portrays women in rural areas as passive victims, highlighting examples where women demonstrate agency in decision-making and farm management, often strategically responding to changes in labor dynamics due to male migration. The findings reveal that women's roles are not static but adapt to changing circumstances, sometimes challenging traditional gender norms. Next, the paper addresses the myth suggesting that the feminization of agriculture is detrimental to agricultural productivity. It argues that productivity comparisons

between men and women fail to consider structural factors such as access to resources and services, which significantly impact productivity. The analysis underscores the importance of addressing inequalities in resource access to enhance women's productivity and contribute to overall agricultural output and food security. Finally, the myth of homogeneity among women farmers is challenged by considering intersectionality, recognizing that women's experiences .The paper advocates for a comprehensive approach to understanding the feminization of agriculture, highlighting the need for data-driven analyses that account for various social identities and their implications for food security.

CHAPTER 3 : OBJECTIVES

- To show female participation rate in the Agricultural sector
- To show the wage disparities faced female by female in the context of
 - Agricultural field labourers
 - Agricultural operations like ploughing ,sowing/transplanting/weeding (STW),and harvesting/threshing/winnowing (HTW)
- To show how Average female wage rate depends on state GDP ,rural female literacy rate and the labour force participation rate of females in rural areas

CHAPTER 4 : DATA AND METHODOLOGY

The paper aims to study the female participation rate in agriculture ,the wage disparities between male and female and the extent of dependence of female wage rate on stategdp , rural female literacy rate and the female labour force participation in rural areas .To examine the study, secondary data is taken into consideration. Secondary data as per the requirement of the study is obtained from various journals, policy papers, published and unpublished records, articles, government websites and reports, database related to women's participation, wage rates.I have relied on quantitative method to carry out this research project by running a regression model and using graphs to help understand the trend.Graphical analysis was done to understand - the participation rate of female from 1881 to 2011 , wage disparities between male and female in the context of annnual wage ratw field laborers from 2010-2020 and various agricultural operations from 2000-2017.To further analyse,I run a regression by taking average female wage rate(dependent variable) on state gdp , rural female literacy rate and female labour force participation of rural areas(independent variables).This data is computed for the year 2011 based on 18 states : Andhra Pradesh ,Assam ,Bihar ,Chhattisgarh ,Gujarat ,Haryana ,Himachal Pradesh ,Jharkhand ,Karnataka ,Kerala ,MadhyaPradesh ,Maharashtra ,Odisha ,Rajasthan ,Tamil Nadu ,Tripura ,Uttar Pradesh and West Bengal.

CHAPTER 5: EMPIRICAL RESULTS AND INTPRETATIONS

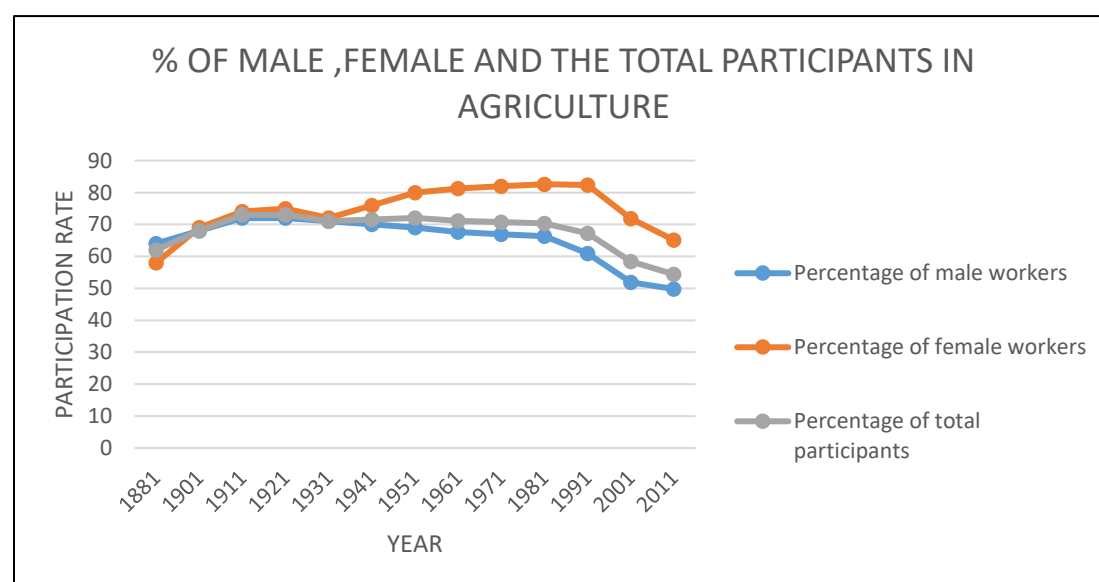
1. To Show female participation rate and wage disparities faced by women in agriculture sector in India.

Table1: Percentage of workers in agriculture (1881-2011)

Year	Percentage of male workers	Percentage of female workers	Percentage of total participants
1881	64	58	62
1901	68	69	68
1911	72	74	73
1921	72	75	73
1931	71	72	71
1941	70	76	71.5
1951	69	80	72
1961	67.65	81.25	71.15
1971	66.97	81.92	70.72
1981	66.3	82.6	70.3
1991	60.9	82.4	67.2
2001	51.9	71.8	58.4
2011	49.8	65.1	54.4

Source : Census of India

FIGURE 1



The data for graph was collected from census of India. This illustrates a decline in the participation rates of both males and females in the agriculture sector. However, the

participation percentage of females is higher than that of males. Additionally, the participation of female workers surpasses the overall participation level, including both male and female. Due to ongoing economic growth and increasing competitions men migrate to city centers for higher wages and employment while women are left with the sole responsibilities of household chores and agricultural activities.

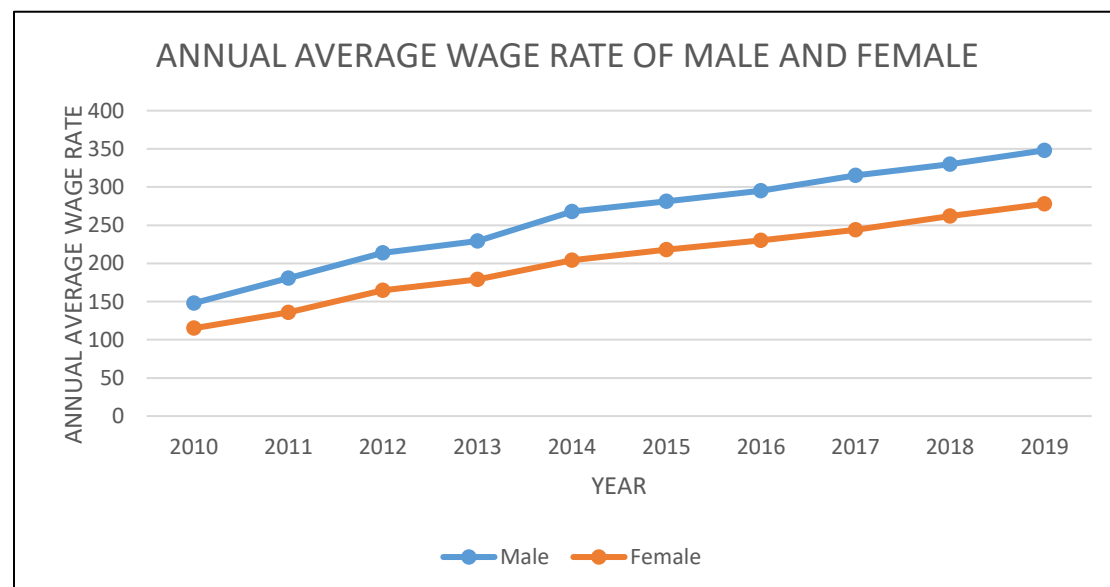
2. To show wage disparities

Table 2. Annual average daily wage rate in (Rs) for field labor

Year	In India	
	Male	Female
2010	147.91	115.02
2011	180.7	135.67
2012	213.71	164.57
2013	229.12	178.82
2014	268	204
2015	281.32	217.76
2016	295	230
2017	315	244
2018	330	262
2019	348	278

Source: Director of economics and statistics department of agriculture, cooperation and farmers welfare

FIGURE 2



The data of the graph is collected from director of economics and statistics department of agriculture, cooperation and farmers welfare, agriculture welfare. The

graph illustrates a significant disparity in the annual average daily wage rates between males and females, where men consistently earned higher wages than females. But the annual daily wage rate for females exhibited a growing trend throughout the reference period (2010-2019).

Table 3. Real wage rate for different agricultural operations in rural India.

Year	Ploughing		STW		HTW	
	Male	Female	Male	Female	Male	Female
2000	113.96	-	93.51	75.37	91.41	75.21
2001	119.04	-	96.31	75.67	93.06	75.93
2002	117.37	-	95.09	75.12	91.95	75.92
2003	117.84	-	96.55	74.29	90.56	75.01
2004	112.01	-	96.05	74.32	90.93	72.48
2005	112.98	-	95.99	74.11	93.53	74.26
2006	111.72	-	94.45	73.16	91.99	73.37
2007	116.14	-	94.53	75.14	93.12	75.33
2008	118.05	-	96.91	78.18	96.88	78.13
2009	120.85	-	98.54	81.71	99.79	82.03
2010	133.66	-	109.36	91.28	108.12	90.1
2011	145.25	-	121.06	102.99	121.24	100.29
2012	156.33	-	128	109.09	129.49	105.96
2013	171.83	-	142.06	117.55	144.77	120.54
2014	175.61	-	154.01	125.16	156.29	132.12
2015	175.56	-	154.49	126.28	155.39	130.21
2016	179.68	-	158.8	131.44	157.12	132.52

Source: Consumer price index for agricultural labourer, Labour bureau.

FIGURE 3

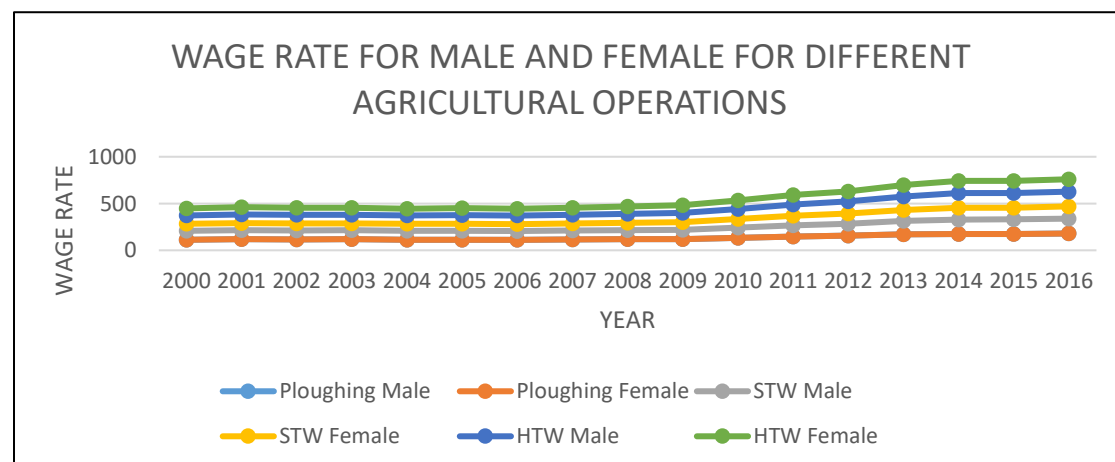


Table 3 presents the real wage rates for various agricultural operations in rural India, including ploughing, sowing/transplanting/weeding (STW) and harvesting/threshing/winnowing (HTW) for the period between 2000 and 2016. Throughout this period, a significant disparity in real wage rates between males and females in different agricultural operations was observed. It is worth noting that no data regarding ploughing by females was observed, possibly due to societal prejudice that perceives females as physically inferior to males. The growth of real wage rates for males remained stagnant or declined from 2000-01 to 2006-07. However, starting from 2007-08, there was a sharp increase in the real wage rates for males, which continued until 2014-15. Interestingly, the wage rates for female occupations exhibited a similar pattern. The real wage rates for females remained stagnant until 2006-07, after which there was a rapid growth in wage rates for females in different agricultural operations from 2007-08 onwards.

Table 4 .Data showing average female wage rate ,stategdp ,rural female literacy rate and rural female labour force participation rate of 18 states in India

STATES	avg female wage rate	stategdp	female literacy rate	LFPR(rural)
Andhra Pradesh	140.71	379402.03	52	44.6
Assam	113.54	143175	63	23.7
Bihar	115.1	247143.9606	49	20.2
Chhattisgarh	87.68	158073.8209	55	46.3
Gujarat	123.22	615606	61	32
Haryana	214.92	297538.5207	60	20.8
Himachal Pradesh	148.16	72719.82932	75	47.4
Jharkhand	99.23	150917.59	49	35
Karnataka	152.2	606009.8108	60	38
Kerala	282.57	364047.8786	91	20
Madhya Pradesh	117.98	315561.59	52	39
Maharashtra	93.13	1280369.438	69	42
Odisha	93.65	230987.0752	61	29.7
Rajasthan	118.41	434837	46	42.7
Tamil Nadu	97.48	751485.7583	65	41.2
Tripura	159.08	359434.11	80	26.3
Uttar Pradesh	131.13	724050	54	18.3
West Bengal	131.59	520485	66	19.4

Table 5 . Regression Results of average female wage rate on stategdp ,rural female literacy rate and rural female labour force participation rate

Residuals:				
Min	1Q	Median	3Q	Max
-49.071	-22.907	-8.848	28.278	64.105
Coefficients:				
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.028e+01	6.330e+01	0.794	0.440
stategdp	-2.870e-05	3.154e-05	-0.910	0.378
literacyrate	2.297e+00	8.082e-01	2.842	0.013 *
`LFPR(rural)`	-1.382e+00	9.027e-01	-1.531	0.148

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				
Residual standard error: 37.78 on 14 degrees of freedom				
Multiple R-squared: 0.4904, Adjusted R-squared: 0.3812				
F-statistic: 4.49 on 3 and 14 DF, p-value: 0.02086				

The coefficient for state GDP is -2.870e-05. This suggests that for each unit increase in state GDP, the average female wage rate decreases by approximately 0.0000287 units. It implies that as the overall economic prosperity of a state increases (reflected in its GDP), the average wage rate for female agricultural workers tends to decrease. This indicate a potential issue of income inequality within the agricultural sector, where despite overall economic growth, wage rates for female workers may not increase proportionally.

The coefficient for female literacy rate is 2.297e+00. This suggests that for each unit increase in female literacy rate of rural areas, the average female wage rate increases by approximately 2.297 units. This highlights the importance of education, particularly for females, in rural areas. Higher levels of female literacy can lead to better skills, awareness of rights, and access to information, which can empower women to demand fairer wages and opportunities within the agricultural sector.

The coefficient for labour force participation rate in rural areas is $-1.382e+00$. This suggests that for each unit increase in labour force participation rate in rural areas, the average female wage rate decreases by approximately 1.382 units. This result might indicate a potential oversupply of labor in rural areas. As more individuals participate in the rural labor force, competition for agricultural jobs increases, leading to a downward pressure on wages for female workers. This could underscore the need for policies and interventions aimed at creating alternative employment opportunities or enhancing productivity within the agricultural sector to mitigate wage depression.

Overall, the model's adjusted R-squared of 0.3812 indicates that around 38.12% of the variation in average female wage rates in agriculture in India can be explained by the three independent variables included in the model (state GDP, literacy rate, and rural labour force participation rate). However, it's essential to consider additional factors like access to land ownership, technological adoption, etc, so these regression results suggest that economic, social, and demographic factors play significant roles in determining female wage rates in the agricultural sector in a country like India.

CHAPTER 6 : DISCUSSION

The dissertation delves into the critical issue of the feminization of the agricultural sector in India and the associated wage disparities faced by female agricultural workers. The introduction provides a comprehensive overview of the historical context, highlighting the significant but often overlooked contributions of women to agriculture. It underscores the need to address gender disparities and empower women in agriculture through policy initiatives that prioritize their participation and well-being.

The empirical analysis presents compelling evidence of the declining participation rates of both male and female workers in the agricultural sector over time. However, it also reveals that the participation of female workers consistently exceeds that of males, indicating the disproportionate burden borne by women in sustaining agricultural livelihoods. Furthermore, the analysis of wage disparities between male and female agricultural workers underscores the existence of significant gender-based inequities in earnings, with men consistently earning higher wages than women. Despite some growth in female wage rates over time, the wage gap remains substantial and persistent.

The regression analysis offers valuable insights into the determinants of average female wage rates in agriculture, highlighting the complex interplay of economic, social, and demographic factors. While higher female literacy rates are associated with higher wage rates, increasing rural labor force participation rates are linked to lower wage rates. These findings suggest the need for targeted interventions to address structural inequalities and create enabling environments for women's economic empowerment in agriculture.

CHAPTER 7 : CONCLUSION AND

RECOMMENDATIONS

In conclusion, the dissertation contributes to our understanding of the challenges and opportunities faced by female agricultural workers in India. It underscores the urgent need for gender-responsive policies and programs that prioritize women's empowerment, improve access to resources, and address wage disparities in the agricultural sector. The research concludes by summarizing the key findings and implications for policy and practice. It highlights highlighting the importance of recognizing and addressing gender disparities in the agricultural workforce and emphasizes the need for targeted interventions to promote women's participation and enhance their economic empowerment. The study underscores the role of education, economic development, and labor market policies in creating an enabling environment for women in agriculture. By recognizing and valuing the contributions of women to agriculture, India can unlock the full potential of its rural workforce and achieve more inclusive and sustainable agricultural development

Recommendations:

- Strengthen efforts to promote female education and literacy in rural areas, particularly focusing on skills development and awareness-raising on women's rights and entitlements.
- Implement gender-sensitive agricultural policies and programs that prioritize women's participation and leadership in decision-making processes, land ownership, access to credit, and extension services.
- Address structural barriers to women's economic empowerment in agriculture, including discriminatory social norms, lack of access to productive resources, and unequal opportunities for skill development and training.
- Enhance monitoring and enforcement of labor laws and regulations to ensure fair and equal wages for male and female agricultural workers, with particular

attention to addressing wage disparities and promoting gender equality in employment practices.

- By implementing these recommendations, India can advance gender equality and social inclusion in agriculture, promote sustainable rural development, and achieve its broader goals of poverty reduction and inclusive growth.

REFERENCES

Bina Agarwal (2008). Women, poverty and agricultural growth in India. The journal of peasant studies, Volume 13, Issue 4, Pages 165 to 220.

Dewan, Ritu. .Contextualising and Visibilising Gender and Work in Rural India: Economic Contribution of Women in Agriculture .Indian Journal of Agricultural Economics; Bombay Vol. 71, Iss. 1, (Jan-Mar 2016): 49-58.

Nitya Chanana-Nag & Pramod K. Aggarwal Woman in agriculture, and climate risks: hotspots for development

Published: 03 August 2018

Volume 158, pages 13–27, (2020)

Nozomi Kawarazuka a, Cheryl R. Doss b,* , Cathy Rozel Farnworth c, Rhiannon Pyburn Myths about the Feminization of agriculture: Implications for global food security Global Food Security Volume 33, June 2022, 100611

Mamata Swain et al. (2019) in the paper Feminisation of Agricultural Labour in India: Issues and Challenges Mahila Pratishtha .ISSN 2454-7891 Volume:4 Issue:3 January – March 2019.

Itishree Pattnaik et.al (2017) in her paper The feminization of agriculture or the feminization of agrarian distress? Tracking the trajectory of women in agriculture in India .Journal of Asia Pacific Economy .Volume 23 Issue 1 Pages 138-155 .

Director of economics and statistics department of agriculture, cooperation and farmers welfare, agriculture welfare.

Consumer price index for agricultural labourer, Labour bureau

Census Of India

Ministry of statistics and program implementation

Carmen Deere (2005) .The Feminization of Agriculture? Economic Restructuring in Rural Latin America.United Nations Research Institute for Social Development (UNRISD) Occasional Paper .ISBN 92-9085-049-3

Ghosh, M. M., & Ghosh , A. (2014). Analysis of Women Participation in Indian Agriculture. IOSR Journal Of Humanities And Social Science (IOSR-JHSS), Volume 19, Issue 5, Ver. IV, 01-06.

Mahata, D., Kumar, A., & Kumar, A. R. (2017). Female Work Force Participation and Women Empowerment in Haryana. ternational Journal of Social, Behavioral, Educational, Economic, Business and Industrial Engineering Vol:11, No:4 , 992-998.

Sen, A. (1999). Development of Freedom. New York: Oxford University Press.

Sudaeshan, R. M., & Bhattacharya, S. (2009). Through the Magnifying Glass: Women's Work and. Economic and Political weekly, 59-66.

Swamikannan, D., & Jeyalakshmi, C. (2015). Women Labour in Agriculture in India: Some Facets. Internatinal Journal of Business and Economic Research,Volume: 1; No: 1, 22-28.